

1 REQUIRED INPUTS

Model the wall section in plan with concrete fibers and actual boundary-zone bar locations.

Concrete	wall polygon vertices (x_p, y_p) and net concrete fibers $(A_{c,i}, x_i, y_i)$
Steel	boundary-zone and distributed bars $(A_{s,j}, x_j, y_j)$
Origin	analysis-model pier origin (x_o, y_o) used for M_x and M_y
Materials	$f'_c, \beta_1, \varepsilon_{cu}, E_s, f_y$
Loads	factored axial load P_u and signed demands M_{ux}, M_{uy}

Order: choose axis/sign $\rightarrow$ apply $P_u \rightarrow$ solve neutral axis depth $c \rightarrow$ recover signed M_x or M_y .

2 COORDINATE SIGN CONVENTION

Compression force is positive. Moments are about the selected pier origin:

$$M_x = F(y - y_o), \quad M_y = -F(x - x_o) \quad (1)$$

M_x	moment about the x axis [lb-in]
F	vertical force in one fiber or bar, positive in compression [lb]
y	y coordinate of force F [in]
y_o	y coordinate of pier origin [in]
M_y	moment about the y axis [lb-in]
x	x coordinate of force F [in]
x_o	x coordinate of pier origin [in]

3 AXIS RUN

Use a unit normal vector pointing toward the compression edge. Run both signs for each axis.

$$\mathbf{n} = \begin{cases} (0, 1) \text{ or } (0, -1) & \text{for } M_x \\ (1, 0) \text{ or } (-1, 0) & \text{for } M_y \end{cases} \quad (2)$$

$\mathbf{n}$ neutral-axis normal vector toward compression edge

$$r_{\max} = \max_p (n_x x_p + n_y y_p) \quad (3)$$

$$d_i = r_{\max} - (n_x x_i + n_y y_i), \quad d_j = r_{\max} - (n_x x_j + n_y y_j)$$

$r_{\max}$	extreme compression-edge coordinate along $\mathbf{n}$ [in]
p	wall polygon vertex index
n_x	x component of $\mathbf{n}$
x_p	x coordinate of wall polygon vertex p [in]
n_y	y component of $\mathbf{n}$
y_p	y coordinate of wall polygon vertex p [in]
d_i	distance from compression edge to concrete fiber i [in]
i	concrete fiber index
x_i	x coordinate of concrete fiber i [in]
y_i	y coordinate of concrete fiber i [in]
d_j	distance from compression edge to steel bar j [in]
j	steel bar index
x_j	x coordinate of steel bar j [in]
y_j	y coordinate of steel bar j [in]

4 FIBER FORCES

$$a = \beta_1 c \quad (4)$$

a	equivalent rectangular compression block depth [in]
β_1	ACI stress-block depth factor
c	neutral-axis depth from extreme compression edge [in]

$$F_{c,i} = \begin{cases} 0.85f'_c A_{c,i} & 0 \leq d_i \leq a \\ 0 & d_i > a \end{cases} \quad (5)$$

$F_{c,i}$ concrete compression force in fiber i [lb]
 f'_c specified concrete compressive strength [psi]
 $A_{c,i}$ net concrete area of fiber i , excluding steel area [in²]

$$\varepsilon_{s,j} = \varepsilon_{cu} \left(1 - \frac{d_j}{c} \right) \quad (6)$$

$\varepsilon_{s,j}$ strain in steel bar j , compression positive
 ε_{cu} ultimate concrete compression strain

$$f_{s,j} = \min(f_y, \max(-f_y, E_s \varepsilon_{s,j})) \quad (7)$$

$f_{s,j}$ stress in steel bar j , compression positive [psi]
 f_y yield strength of longitudinal reinforcement [psi]
 E_s modulus of elasticity of steel [psi]

$$F_{s,j} = f_{s,j} A_{s,j} \quad (8)$$

$F_{s,j}$ force in steel bar j , compression positive [lb]
 $A_{s,j}$ area of steel bar j [in²]

5 AXIAL LOAD SETS NEUTRAL AXIS

P_u is in the neutral-axis solve; moment strength is recovered after axial equilibrium is satisfied.

$$P_n(c, \mathbf{n}) = \sum_i F_{c,i} + \sum_j F_{s,j} \quad (9)$$

P_n nominal axial strength for selected c and $\mathbf{n}$ [lb]

$$\varepsilon_t = \max\left(0, \max_j(-\varepsilon_{s,j})\right), \quad \varepsilon_y = \frac{f_y}{E_s} \quad (10)$$

ε_t largest tensile strain in steel, reported positive
 ε_y steel yield strain

$$\phi = \begin{cases} 0.65 & \varepsilon_t \leq \varepsilon_y \\ 0.65 + 0.25 \frac{\varepsilon_t - \varepsilon_y}{0.005 - \varepsilon_y} & \varepsilon_y < \varepsilon_t < 0.005 \\ 0.90 & \varepsilon_t \geq 0.005 \end{cases} \quad (11)$$

ϕ strength reduction factor for tied wall boundary reinforcement

$$R_P(c, \mathbf{n}) = \phi(c) P_n(c, \mathbf{n}) - P_u = 0 \quad (12)$$

R_P axial residual for neutral-axis iteration [lb]
 P_u factored axial load, compression positive [lb]

CONTROL

Compression $P_u > 0$	generally increases c and shifts N.A. toward the tension side
Tension $P_u < 0$	generally decreases c and shifts N.A. toward the compression side
No $R_P = 0$ solution	axial demand outside available section strength for that axis run

6 MOMENT STRENGTH

$$M_{nx}(c, \mathbf{n}) = \sum_i F_{c,i}(y_i - y_o) + \sum_j F_{s,j}(y_j - y_o) \quad (13)$$

M_{nx} nominal moment strength about x axis for selected axis run [lb-in]

$$M_{ny}(c, \mathbf{n}) = - \sum_i F_{c,i}(x_i - x_o) - \sum_j F_{s,j}(x_j - x_o) \quad (14)$$

M_{ny} nominal moment strength about y axis for selected axis run [lb-in]

$$M_{\phi x} = \phi M_{nx}, \quad M_{\phi y} = \phi M_{ny} \quad (15)$$

$M_{\phi x}$ signed design moment strength about x axis [lb-in]

$M_{\phi y}$ signed design moment strength about y axis [lb-in]

7 STRENGTH CHECK

Check the matching axis and sign.

$$|M_{\phi x}^{\pm}| \geq |M_{ux}^{\pm}|, \quad |M_{\phi y}^{\pm}| \geq |M_{uy}^{\pm}| \quad (16)$$

$M_{\phi x}^{\pm}$ design x -axis strength for matching positive or negative demand [lb-in]

$M_{ux}^{\pm}$ factored x -axis moment demand, positive or negative [lb-in]

$M_{\phi y}^{\pm}$ design y -axis strength for matching positive or negative demand [lb-in]

$M_{uy}^{\pm}$ factored y -axis moment demand, positive or negative [lb-in]

CHECK ORDER

- 1 Use the same (x_o, y_o) and sign convention as the analysis model.
- 2 Mesh the actual wall shape; L and U walls do not need rectangular simplification.
- 3 Place boundary-zone bars at actual (x_j, y_j) coordinates.
- 4 Run M_x^+ , M_x^- , M_y^+ , and M_y^- separately.
- 5 For each run, iterate c until $R_P = 0$ using the factored axial load P_u .
- 6 Compute signed $M_{\phi x}$ or $M_{\phi y}$ and compare only to the matching demand sign.